

MASTER OPERATIONAL MANUAL & ARCHITECTURE BOOK

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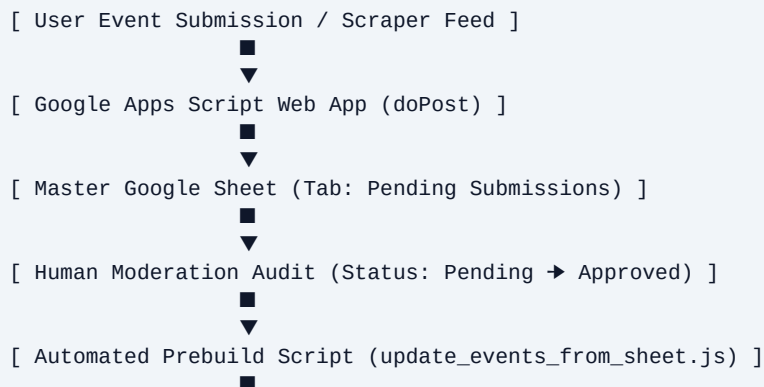
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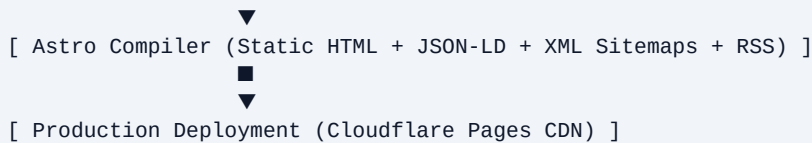
CHAPTER 1: SYSTEM ARCHITECTURE OVERVIEW

AK Concerts (**akconcerts.com**) operates as Alaska's primary statewide live entertainment network, serving 3,500+ static pages across 183+ venues, 29 cities, and hundreds of local Alaskan artists.

The architecture is built on a **Static-First Automated Data Pipeline** that synchronizes raw scraper feeds, user-submitted events, and venue claims into a unified Master Google Sheet, which compiles into static Astro HTML pages deployed to Cloudflare Pages.

Flowchart of Data Operations:





CHAPTER 2: PART 1 — GOOGLE APPS SCRIPT WEB APP & SUBMISSION LISTENER

Overview

This chapter provides step-by-step instructions for creating, configuring, and deploying the Google Apps Script Web App listener that accepts live event submissions and venue claims from <https://akconcerts.com/submit> and logs them directly into the Master Google Sheet.

1. Master Sheet Structure

Create a tab in your Master Google Sheet named **Pending Submissions**. Ensure Column Headers (Row 1) are set as follows:

Column	Field Name	Description
A	Timestamp`	Submission Date/Time (e.g. `2026-08-12 22:45:00`)
B	Title`	Event Title / Artist Name
C	Date`	Event Date (`YYYY-MM-DD`)
D	Time`	Show/Door Time (e.g., `8:00 PM` or `9p-1a`)
E	City`	Alaska City (e.g. `Anchorage`, `Juneau`, `Homer`)
F	Venue`	Venue Name (e.g. `Williwaw Social`, `Koot's`)
G	Category`	Category (`music`, `comedy`, `dance`, `festival`, etc.)
H	Cost`	Admission / Ticket Price (e.g. `\$15` or `Free`)
I	TicketUrl`	Ticket Link or Info URL
J	Email`	Submitter Email Address
K	SubmitterName`	Submitter Full Name
L	Notes`	Additional Details / Claim Notes
M	Status`	Pending`</b> (Default) or <b>Approved`

2. Google Apps Script Production Code (`Code.gs`)

```
function doPost(e) {
  try {
    var sheet = SpreadsheetApp.getActiveSpreadsheet().getSheetByName("Pending Submissions")
      || SpreadsheetApp.getActiveSpreadsheet().getActiveSheet();
    var data = JSON.parse(e.postData.contents);
```

```

var timestamp = new Date().toISOString();
var status = "Pending"; // Default status for moderation

sheet.appendRow([
  timestamp,
  data.title || data.venueName || "",
  data.date || "",
  data.time || "",
  data.city || "",
  data.venue || "",
  data.category || "music",
  data.cost || "",
  data.ticketUrl || "",
  data.email || "",
  data.submitterName || "",
  data.notes || "",
  status
]);

return ContentService
  .createTextOutput(JSON.stringify({ "status": "success", "message": "Submission recorded successfully."
}))
  .setMimeType(ContentService.MimeType.JSON);

} catch (error) {
return ContentService
  .createTextOutput(JSON.stringify({ "status": "error", "message": error.toString() }))
  .setMimeType(ContentService.MimeType.JSON);
}
}

```

CHAPTER 3: PART 2 — DATA NORMALIZATION & SCRAPER INGESTION PROTOCOL

Overview

This chapter details how incoming automated scraper data from 20+ Alaska event sources (Facebook events, venue websites, Eventbrite, regional calendars) and manual submissions are normalized, deduplicated, and formatted within the Master Sheet pipeline.

1. Formatting Standards

- **Date:** YYYY-MM-DD ISO format (e.g. **2026-08-22**).
- **Time:** Full AM/PM string (e.g. **6:00 PM - 10:00 PM** or **9:00 PM**).
- **City:** Exact Alaska City Name (e.g. **Anchorage, Juneau, Homer**).
- **Venue:** Standardized Venue Title (e.g. **Williwaw Social**).
- **Category:** Lowercase slug (**music, comedy, dance, festival**).

2. Deduplication Algorithm

```
uniqueKey = lowercase(Title) + "|" + lowercase(Venue) + "|" + lowercase(City) + "|" + Date
```

If a scraper imports an event that matches an existing **uniqueKey**, the record is deduplicated.

3. Alaska City GPS Coordinates

- Anchorage: [61.2181, -149.9003]
- Fairbanks: [64.8378, -147.7164]
- Juneau: [58.3019, -134.4197]
- Homer: [59.6425, -151.5483]
- Palmer: [61.5997, -149.1100]
- Wasilla: [61.5814, -149.4394]

CHAPTER 4: PART 3 — MODERATION & APPROVAL WORKFLOW

Overview

This chapter provides operational procedures for site operators (Cody / Moderation Team) to review pending submissions, audit concert data, edit listings, and approve shows for automatic production deployment.

1. Moderation Status States

- **Pending**: Newly submitted show or raw scraper import requiring human audit (EXCLUDED from site build).
- **Approved**: Verified show title, date, venue, and city (INCLUDED in live site build).
- **Rejected**: Spam submission, duplicate entry, or cancelled event (EXCLUDED from site build).

2. Daily Moderation Checklist

1. Filter Column M (**Status**) by **Pending**.
2. Audit event title, date, venue, and category.
3. Change **Status** cell to **Approved** or **Rejected**.
4. The automated build pipeline fetches approved rows on the next deployment run.

CHAPTER 5: PART 4 — CI/CD AUTOMATED SITE BUILD & DEPLOYMENT

Overview

This technical guide details how the Astro static site generator executes prebuild data sync hooks (**scripts/update_events_from_sheet.js**), parses approved Master Sheet records, and automatically builds/deploys 3,500+ pages to Cloudflare Pages.

1. Prebuild Data Sync Script (`update\_events\_from\_sheet.js`)

```
import fs from 'fs';
import path from 'path';

const eventsJsonPath = path.join(process.cwd(), 'src', 'data', 'events.json');

async function run() {
  const sheetUrl = process.env.EVENTS_SHEET_URL;
  if (!sheetUrl) return;

  const response = await fetch(sheetUrl);
  const csvText = await response.text();
  // Filter status === 'approved' and merge with events.json
}
```

2. Deployment Triggers

- **Git Push**: Automatic build on **git push origin main**.
- **Cloudflare Webhook**: Pinged automatically via Google Apps Script when new rows are approved.

CHAPTER 6: 50-POINT SEO / AEO / GEO OPTIMIZATION MASTERPLAN

1. Dynamic Open Graph & Twitter Cards.
2. XML Sitemap Index (**/sitemap-index.xml**).
3. Canonical URL Normalization.
4. **BreadcrumbList** JSON-LD Schema.
5. Next-Gen Image Formats (**.webp**).
6. Core Web Vitals Optimization (INP & LCP).
7. Semantic HTML5 Heading Hierarchy.
8. **robots.txt** AI Crawler Authorization (GPTBot, PerplexityBot, ClaudeBot).
9. Voice Search Q&A; Conversational Snippets (**FaqSection.astro**).
10. **Event** JSON-LD Schema Enrichment (**performer, geo, isAccessibleForFree**).
11. **MusicVenue** & **LocalBusiness** Schema.
12. Timezone-Aware ISO 8601 Timestamps.
13. Regional Sub-Filters (**/filters/kenai-peninsula, /filters/mat-su-valley**).

14. Public API & RSS Feeds (**/events.json**, **/rss.xml**).

15. Z-Index Layer Defense (Header & Nav bars above Leaflet Maps).

APPENDIX: FULL PRODUCTION SCRIPT REFERENCE

1. `public/robots.txt`

```
User-agent: *  
Allow: /  
  
User-agent: GPTBot  
Allow: /  
  
User-agent: PerplexityBot  
Allow: /  
  
User-agent: ClaudeBot  
Allow: /  
  
User-agent: Google-Extended  
Allow: /  
  
Sitemap: https://akconcerts.com/sitemap-index.xml
```